

PRESENTER

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Abstract :

- Statistics underpin trustworthy research. Our audit of 23,000 biomedical papers found only 4.49% had automatically checkable p-values.
- Missing test statistics and degrees of freedom hinder verification.
- StickForStats supports reproducible analysis through automated assumption checks, guided test selection and recorded workflows.

Background :

Why do we need statistics?

Large Language Models work on statistics:

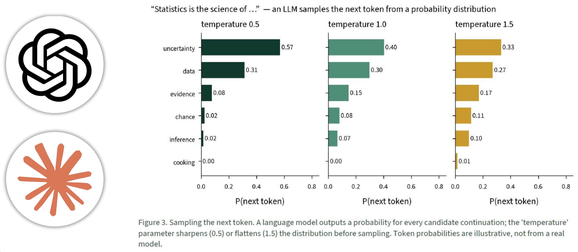
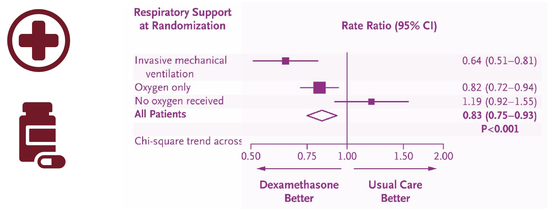


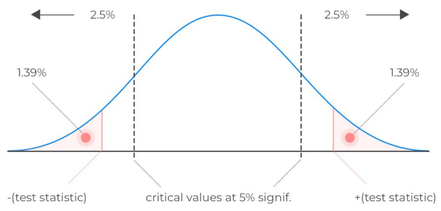
Figure 3. Sampling the next token. A language model outputs a probability for every candidate continuation; the 'temperature' parameter sharpens (0.5) or flattens (1.5) the distribution before sampling. Token probabilities are illustrative, not from a real model.

Statistics identifies evidence of dexamethasone for improved survival in ventilated COVID-19 patients



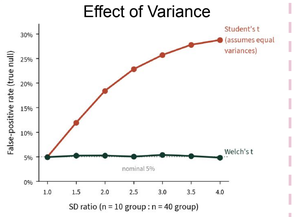
What does a complete test result look like?

p-value is the likelihood of an event; **t-statistic** is the signal over noise; **df:** degrees of freedom depends on sample size



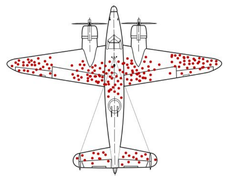
Consequences of wrong statistics!

Effect of Variance



Unchecked variance can lead to false positives

Survivorship bias



Negative results not reported in scientific publications

Sally Clark Case 1999

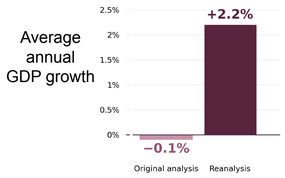


The chance of sudden infant death syndrome (SIDS) is 1 in 8543

Wrong independent assumption $= \left(\frac{1}{8543}\right)^2$

= 1 in 73 million

Reinhart–Rogoff debt-and-growth study Case 2010



Excluding observations and wrong weighting of averages suggested lower GDP for high-debt group countries

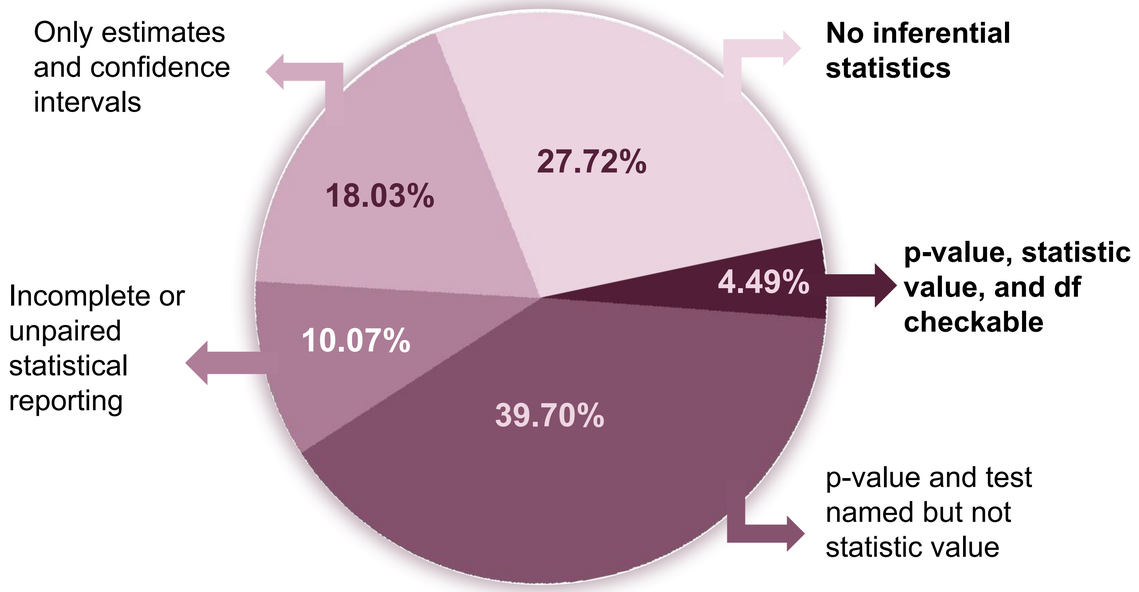
Most biomedical articles print no statistical result a reader can recompute



Audit of randomly selected 23,000 open-access PMC articles (2020 to 2025)



Looking through text about information on test statistic-value and degrees of freedom (df) to recalculate p-value



StickForStats for guided and automated statistical analysis



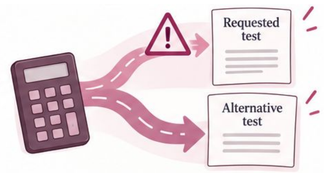
Look Here!!



StickForStats



Checks the data as per the experiment



Guardian system selects the right statistical test



Recorded analysis for reproducibility

Free, open-sourced software for statistical analysis

Take Home message: Print the test statistic, its df and the exact p for every test, and let the reader judge significance

Scan here for more info on:
StickForStats software
Interesting stories involving statistics

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